

Program 2 :

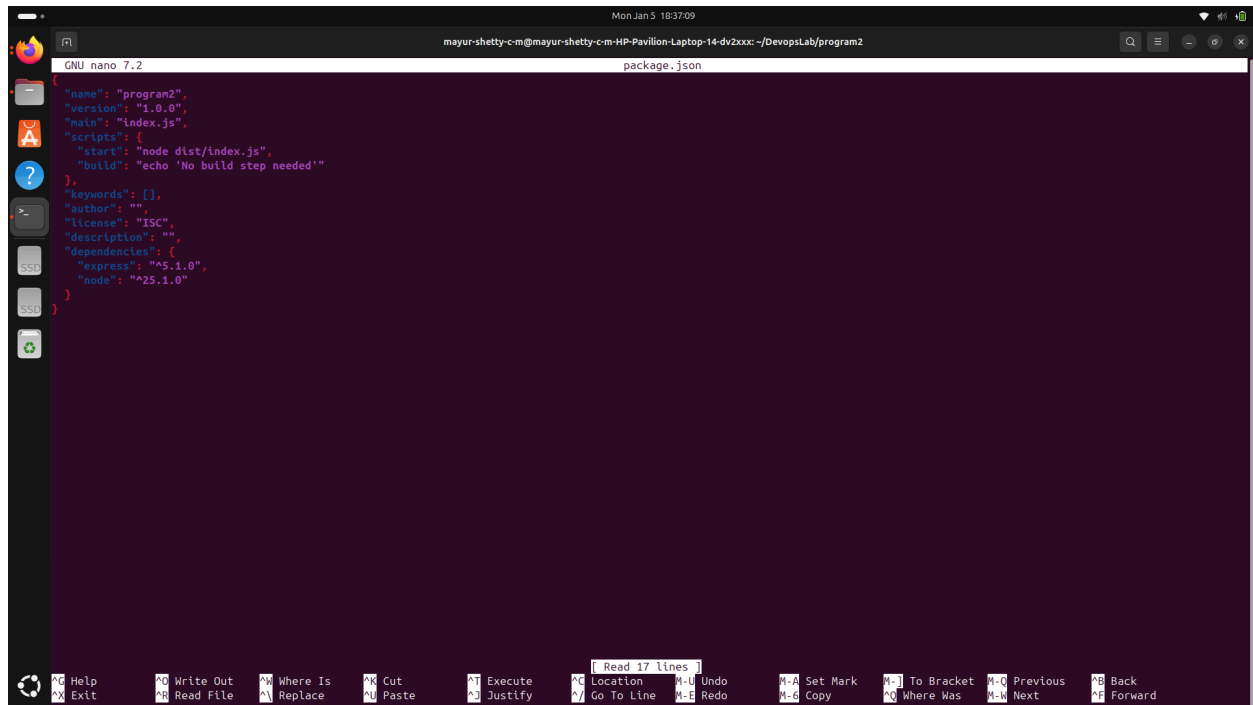
Develop a Multi-Stage Dockerfile for Container Orchestration.

-Dockerfile

- [index.js](#)

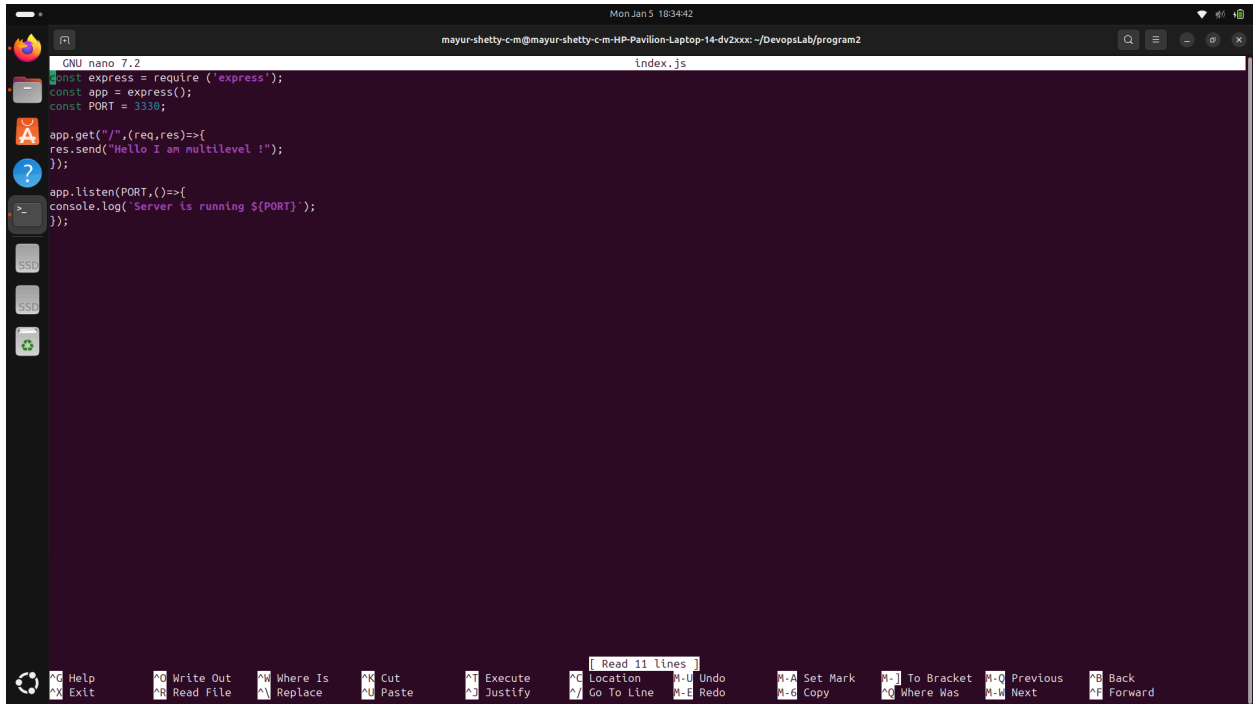
Step 1: Create a package.json

nano package.json



```
GNU nano 7.2 package.json
{
  "name": "program2",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "start": "node dist/index.js",
    "build": "echo 'No build step needed'"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": "",
  "dependencies": {
    "express": "^5.1.0",
    "node": "^25.1.0"
  }
}
```

Step2: Create [index.js](#)

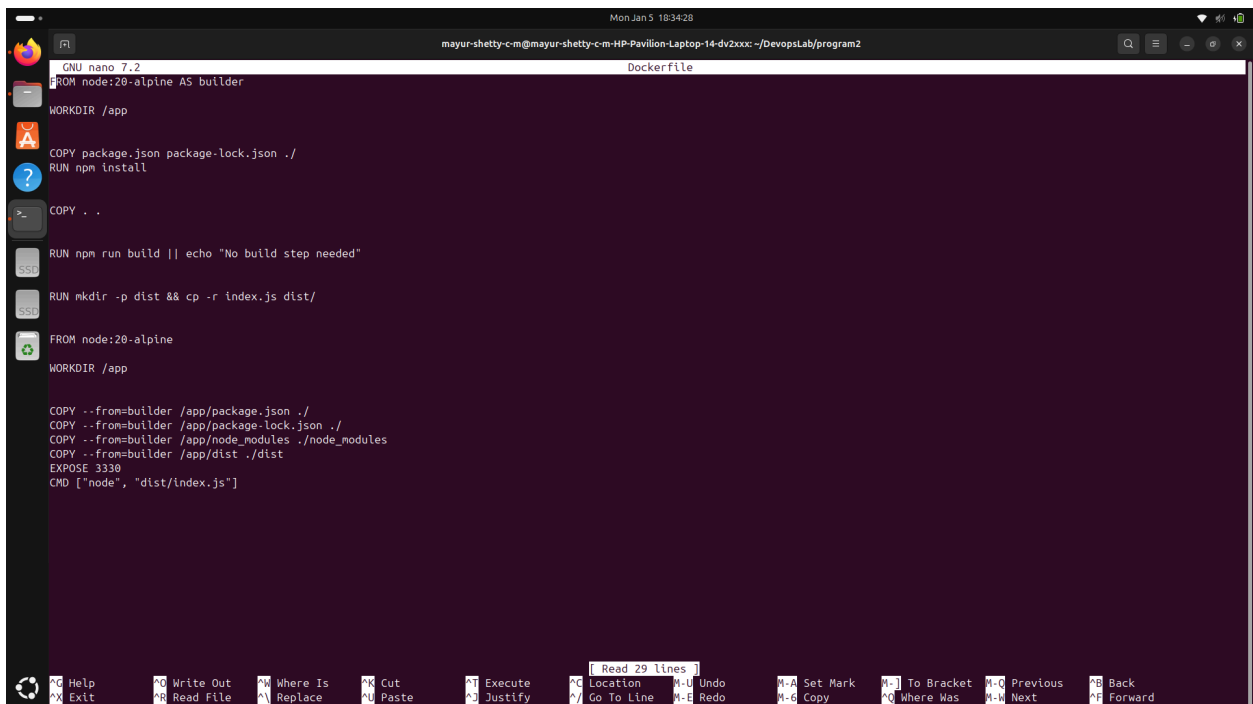


```
GNU nano 7.2 index.js
const express = require('express');
const app = express();
const PORT = 3330;

app.get("/",(req,res)=>{
  res.send("Hello I am multilevel !");
});

app.listen(PORT,()=>{
  console.log(`Server is running ${PORT}`);
});
```

Step 3 : Create a Docker file



```
FROM node:20-alpine AS builder
WORKDIR /app
COPY package.json package-lock.json ./
RUN npm install
COPY . .
RUN npm run build || echo "No build step needed"
RUN mkdir -p dist && cp -r index.js dist/
FROM node:20-alpine
WORKDIR /app
COPY --from=builder /app/package.json ./
COPY --from=builder /app/package-lock.json ./
COPY --from=builder /app/node_modules ./node_modules
COPY --from=builder /app/dist ./dist
EXPOSE 3330
CMD ["node", "dist/index.js"]
```

Step 4 : Install the files

npm install express

npm install

Npm init -y

Step 4 : Build the docker file

Docker build -t program2 .

```
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~/DevopsLab/program2$ docker build -t program2 .
[+] Building -19660.4s (17/17) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 504B
=> [internal] load metadata for docker.io/library/node:20-alpine
=> [auth] library/node:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [builder 1/7] FROM docker.io/library/node:20-alpine@sha256:658d0f63e501824d6c23e06d4bb95c71e7d704537c9d9272f488
=> => resolve docker.io/library/node:20-alpine@sha256:658d0f63e501824d6c23e06d4bb95c71e7d704537c9d9272f488ac03a370
=> => sha256:fcbb8f7d018707c656a4da2eea8a15f2893d2258093fea9ff2ea5ea1c8a82112 1.72kB / 1.72kB
=> => sha256:e80b0510ba947015cacdddea3d23dc761e399971c074a1bf32eea7e44510524 6.52kB / 6.52kB
=> => sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612 3.86MB / 3.86MB
=> => sha256:8d06ba6946d1f299d8f962e37906c77006df33d47050430a57f7893ec35af697 42.78MB / 42.78MB
=> => sha256:cb3325e64457574e24f92246e3da3376946e473d636209e1390eac47b50b26a3 1.26MB / 1.26MB
=> => sha256:658d0f63e501824d6c23e06d4bb95c71e7d704537c9d9272f488ac03a370d448 7.67kB / 7.67kB
=> => sha256:fd1849a5c548bc65ee47a64498951bda5d40e87d08efe9dca69b5c8cdedc7a52 443B / 443B
=> => extracting sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612
=> => extracting sha256:8d06ba6946d1f299d8f962e37906c77006df33d47050430a57f7893ec35af697
=> => extracting sha256:cb3325e64457574e24f92246e3da3376946e473d636209e1390eac47b50b26a3
=> => extracting sha256:fd1849a5c548bc65ee47a64498951bda5d40e87d08efe9dca69b5c8cdedc7a52
=> [internal] load build context
=> => transferring context: 188.08MB
=> [builder 2/7] WORKDIR /app
=> [builder 3/7] COPY package.json package-lock.json ./
=> [builder 4/7] RUN npm install
=> [builder 5/7] COPY . .
=> [builder 6/7] RUN npm run build || echo "No build step needed"
=> [builder 7/7] RUN mkdir -p dist && cp -r index.js dist/
=> [stage-1 3/6] COPY --from=builder /app/package.json ./
=> [stage-1 4/6] COPY --from=builder /app/package-lock.json ./
=> [stage-1 5/6] COPY --from=builder /app/node_modules ./node_modules
=> [stage-1 6/6] COPY --from=builder /app/dist ./dist
=> exporting to image
=> => exporting layers
=> => writing image sha256:7a220b579a34df00f19ca22c49f5b6f4a211834ca3ad3e7f5f6b7a5911bf32c0
=> => naming to docker.io/library/program2
mayur-shetty-c-m@mayur-shetty-c-m-HP-Pavilion-Laptop-14-dv2xxx:~/DevopsLab/program2$
```

Step 5 : Run in the port with the port binding method

docker run -d -p 3330:3330 program2



Hello I am multilevel !